ods层建表语句

```
create table logs_cwy_ods(
    line string
) partitioned by (dt string)
stored as
    inputformat "com.hadoop.mapred.DeprecatedLzoTextInputFormat"
outputformat "org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat"
-- 指定存储位置 注意修改成你的路径，放到我这里，等会搞乱了
location '/cwy_102/warehouse/logs/logs_cwy/logs_cwy_ods';
```

```
# 每日装载脚本
#!/bin/bash

# 定义变量方便修改 库名
APP=cwy_102

# 如果是输入的日期按照取输入日期；如果没输入日期取当前时间的前一天
if [ -n "$1" ] ;then
   do_date=$1
else
   do_date=`date -d "-1 day" +%F`
fi

echo ================== 日志日期为 $do_date ==================
sql="
load data inpath '/cwy_102/logs/log_cwy/$do_date' into table ${APP}.logs_cwy_ods
partition(dt='$do_date');
"
hive -e "$sql"

#添加 lzo压缩索引
hadoop jar /opt/module/hadoop/share/hadoop/common/hadoop-lzo-0.4.20.jar
com.hadoop.compression.lzo.DistributedLzoIndexer
/cwy_102/warehouse/logs/logs_cwy/logs_cwy_ods/dt=$do_date
```

dwd层

```
create table logs_cwy_dwd(
    ip STRING,
    date_time STRING,
    method STRING,
    url STRING,
    status INT,
    size BIGINT,
    response_time STRING
) comment "页面访问日志表"
partitioned by (dt string)
STORED AS PARQUET
-- 指定存储位置 注意修改成你的路径，放到我这里，等会搞乱了
LOCATION "/cwy_102/warehouse/logs/logs_cwy/logs_cwy_dwd"
```

```
tblproperties ('parquet.compression'='lzo');
```

```
# 首日装载脚本  第一天执行一次
#!/bin/bash

# 定义变量方便修改 库名
APP=cwy_102

if [ -n "$2" ] ;then
   do_date=$2
else
   echo "请传入日期参数"
   exit
fi

logs_cwy_dwd="
set hive.exec.dynamic.partition.mode=nonstrict;
set hive.input.format=org.apache.hadoop.hive.ql.io.HiveInputFormat;
insert overwrite table ${APP}.logs_cwy_dwd partition (dt)
select
    data[0] as ip,
    data[1] as date_time,
    data[2] as method,
    data[3] as url,
    data[4] as status,
    data[5] as size,
    data[6] as response_time,
    date_format(data[1],"yyyy-MM-dd") as dt
from
    (
        select
             split(line,'  ') as data
        from ${APP}.logs_cwy_ods
        where dt='$do_date'
    ) o1;
"

case $1 in
    logs_cwy_dwd )
        hive -e "$logs_cwy_dwd"
    ;;
    all )
        hive -e "$logs_cwy_dwd"
    ;;
esac
```

```
# 每日装载
#!/bin/bash

APP=cwy_102
# 如果是输入的日期按照取输入日期；如果没输入日期取当前时间的前一天
if [ -n "$2" ] ;then
    do_date=$2
else
    do_date=`date -d "-1 day" +%F`
```

```
fi

logs_cwy_dwd="
set hive.input.format=org.apache.hadoop.hive.ql.io.HiveInputFormat;
insert overwrite table ${APP}.logs_cwy_dwd partition (dt='$do_date')
select
    data[0] as ip,
    data[1] as date_time,
    data[2] as method,
    data[3] as url,
    data[4] as status,
    data[5] as size,
    data[6] as response_time
from
(
         select
              split(line,'  ') as data
         from ${APP}.logs_cwy_ods
         where dt='$do_date'
) o1;
"
case $1 in
    logs_cwy_dwd )
        hive -e "$logs_cwy_dwd"
    ;;
    all )
        hive -e "$logs_cwy_dwd"
    ;;
esac
```

dws层

```
-- 统计每日数据临时表
create table dws_count_day1(
                            url STRING,
                            ip STRING,
                            cnt bigint,
                            max_response_time STRING,
                            min_response_time STRING,
                            max_size BIGINT,
                            min_size BIGINT
)comment "每日统计表临时表"
    partitioned by (dt STRING)
    stored as parquet
    -- 指定存储位置 注意修改成你的路径，放到我这里，等会搞乱了
    location "/cwy_102/warehouse/logs/logs_cwy/dws_count_day1"
    tblproperties ("parquet.compression"="lzo");



-- 统计每日uv表
create table dws_count_uv(
                            ip STRING,
                            ip_cnt bigint,
                            url_cnt bigint,
```

```
                                max_response_time STRING,
                                min_response_time STRING
)comment "每日uv统计表"
    partitioned by (dt STRING)
    stored as parquet
       -- 指定存储位置 注意修改成你的路径，放到我这里，等会搞乱了
    location "/cwy_102/warehouse/logs/logs_cwy/dws_count_uv"
    tblproperties ("parquet.compression"="lzo");
```

```
# 首日装宅
#!/bin/bash
APP=cwy_102


if [ -n "$2" ] ;then
    do_date=$2
else
    do_date=`date -d "-1 day" +%F`
fi

dws_count_day1="
set hive.input.format=org.apache.hadoop.hive.ql.io.HiveInputFormat;
insert overwrite table ${APP}.dws_count_day1 partition  (dt='$do_date')
select url,
       ip,
       count(1) as cnt, -- 访问次数
       max(response_time) as max_response_time,min(response_time) as
min_response_time,
       max(size) as max_size,min(size) as min_size
from ${APP}.logs_cwy_dwd
where dt='$do_date'
group by url,ip;
"

dws_count_uv="
set hive.input.format=org.apache.hadoop.hive.ql.io.HiveInputFormat;
insert overwrite table ${APP}.dws_count_uv partition (dt='$do_date')
select
ip,sum(cnt) as ip_cnt,count(url) url_cnt,max(max_response_time)
max_response_time,
min(min_response_time) min_response_time
from ${APP}.dws_count_day1
where dt='$do_date'
group by ip;
"

case $1 in
    dws_count_day1 )
        hive -e "$dws_count_day1"
    ;;
    dws_count_uv )
        hive -e "$dws_count_uv"
    ;;
    all )
        hive -e "$dws_count_day1$dws_count_uv"
```

```
    ;;
esac
```

```
# 首日装宅
#!/bin/bash
APP=cwy_102
# 如果是输入的日期按照取输入日期；如果没输入日期取当前时间的前一天

if [ -n "$2" ] ;then
   do_date=$2
else
   echo "请传入日期参数"
   exit
fi

dws_count_day1="
set hive.exec.dynamic.partition.mode=nonstrict;
set hive.input.format=org.apache.hadoop.hive.ql.io.HiveInputFormat;
insert overwrite table ${APP}.dws_count_day1 partition (dt)
select url,
       ip,
       count(1) as cnt, -- 访问次数
       max(response_time) as max_response_time,min(response_time) as
min_response_time,
       max(size) as max_size,min(size) as min_size,
       dt
from ${APP}.logs_cwy_dwd
where dt='$do_date'
group by url,ip;
"

dws_count_uv="
set hive.exec.dynamic.partition.mode=nonstrict;
set hive.input.format=org.apache.hadoop.hive.ql.io.HiveInputFormat;
insert overwrite table ${APP}.dws_count_uv partition (dt)
select
ip,sum(cnt) as ip_cnt,count(url) url_cnt,max(max_response_time)
max_response_time,
min(min_response_time) min_response_time,dt
from ${APP}.dws_count_day1
where dt='$do_date'
group by ip;
"

case $1 in
    dws_count_day1 )
        hive -e "$dws_count_day1"
    ;;
    dws_count_uv )
        hive -e "$dws_count_uv"
    ;;
    all )
        hive -e "$dws_count_day1$dws_count_uv"
    ;;
esac
```

ads层

```sql
-- 需求1 每个ip，当天,最近三天，最近7天的访问次数,
-- ip最快响应时长,最慢响应时长
create table ads_ipCount_day3and7(
    dt STRING comment "统计日期",
    ip STRING,
    day1_ip_cnt BIGINT,
    day1_url_cnt BIGINT,
    day3_ip_cnt BIGINT,
    day3_url_cnt BIGINT,
    day7_ip_cnt BIGINT,
    day7_url_cnt BIGINT
) comment "每个ip当天,最近3天,7天的访问次数"
row format delimited fields terminated by "\t"
    -- 指定存储位置 注意修改成你的路径，放到我这里，等会搞乱了
location "/cwy_102/warehouse/logs/logs_cwy/ads_ipCount_day3and7";


-- 需求2 连续3天访问的ip，以及连续天数,访问其实日期，访问结束日期
create table ads_3continuity_request_ip(
    dt STRING comment "统计日期",
    ip STRING comment "ip",
    v_cnt_day bigint comment "连续访问天数",
    start_date STRING comment "起始时间",
    end_date STRING comment "结束时间"
) comment "连续3天访问的用户"
row format delimited fields terminated by "\t"
    -- 指定存储位置 注意修改成你的路径，放到我这里，等会搞乱了
location "/cwy_102/warehouse/logs/logs_cwy/ads_3continuity_request_ip";
```

```bash
# 每日装宅
#!/bin/bash
APP=cwy_102


if [ -n "$2" ] ;then
    do_date=$2
else
    do_date=`date -d "-1 day" +%F`
fi

ads_ipCount_day3and7="
insert overwrite table ${APP}.ads_ipCount_day3and7
select
    '$do_date' as dt,
    ip,
    sum(if(dt>date_add('$do_date',-1),ip_cnt,0))  as day1_ip_cnt,
    sum(if(dt>date_add('$do_date',-1),url_cnt,0))  as day1_url_cnt,
    sum(if(dt>date_add('$do_date',-2),ip_cnt,0)) as day3_ip_cnt,
    sum(if(dt>date_add('$do_date',-2),url_cnt,0)) as day3_url_cnt,
    sum(if(dt>date_add('$do_date',-6),ip_cnt,0)) as day7_ip_cnt,
    sum(if(dt>date_add('$do_date',-6),url_cnt,0)) as day7_url_cnt
from ${APP}.dws_count_uv
where dt>date_add('$do_date',-6)
group by ip;
```

```
"

ads_3continuity_request_ip="
insert overwrite table ${APP}.ads_3continuity_request_ip
select
    '$do_date' as dt,
    ip,count(1) as v_cnt_day,date_diff as start_date,max(dt) as end_date
from (
         select ip,dt,date_add(dt,-row_number() over (partition by ip order by
dt)) as date_diff
         from ${APP}.dws_count_day1
         group by ip,dt
     ) o1
group by  ip,date_diff
 -- having count(1) >=3 ;
"

case $1 in
    ads_3continuity_request_ip )
        hive -e "$ads_ipCount_day3and7"
    ;;
    ads_3continuity_request_ip )
        hive -e "$ads_3continuity_request_ip"
    ;;
    all )
        hive -e "$ads_ipCount_day3and7$ads_3continuity_request_ip"
    ;;
esac
```

Mysql报表

```
-- 字段类型，顺序，个数 要和ads层一样对应，不然会导出不成功
CREATE TABLE `ads_3continuity_request_ip` (
  `dt` varchar(255) NOT NULL,
  `ip` varchar(255) NOT NULL,
  `v_cnt_day` bigint(20) DEFAULT NULL,
  `start_date` varchar(255) DEFAULT NULL,
  `end_date` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`ip`,`dt`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE `ads_ipCount_day3and7` (
  `dt` varchar(255) NOT NULL,
  `ip` varchar(255) NOT NULL,
  `day1_ip_cnt` bigint(20) DEFAULT NULL,
  `day1_url_cnt` bigint(20) DEFAULT NULL,
  `day3_ip_cnt` bigint(20) DEFAULT NULL,
  `day3_url_cnt` bigint(20) DEFAULT NULL,
  `day7_ip_cnt` bigint(20) DEFAULT NULL,
  `day7_url_cnt` bigint(20) DEFAULT NULL,
  PRIMARY KEY (`ip`,`dt`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```bash
#!/bin/bash

hive_db_name=cwy_102
mysql_db_name=cwy_102

export_data() {
/opt/module/sqoop/bin/sqoop export \
--connect "jdbc:mysql://10.32.30.250:3306/${mysql_db_name}?
useUnicode=true&characterEncoding=utf-8"  \
--username root \
--password root \
--table $1 \
--num-mappers 1 \
--export-dir /cwy_102/warehouse/logs/logs_cwy/$1 \
--input-fields-terminated-by "\t" \
--update-mode allowinsert \
--update-key $2 \
--input-null-string '\\N'    \
--input-null-non-string '\\N'
}

# 调用函数
case $1 in
  "ads_ipCount_day3and7" )
    export_data "ads_ipCount_day3and7" "dt,ip"
  ;;
  "ads_3continuity_request_ip")
    export_data "ads_3continuity_request_ip" "dt,ip"
  ;;
  "all" )
    export_data "ads_ipCount_day3and7" "dt,ip"
    export_data "ads_3continuity_request_ip" "dt,ip"
  ;;
esac
```

```bash
/opt/module/sqoop/bin/sqoop export \
--connect "jdbc:mysql://10.32.30.250:3306/cwy_102?
useUnicode=true&characterEncoding=utf-8"  \
--username root \
--password root \
--table "ads_ipCount_day3and7" \
--num-mappers 1 \
--export-dir /cwy_102/warehouse/logs/logs_cwy/ads_ipCount_day3and7 \
--input-fields-terminated-by "\t" \
--update-mode allowinsert \
--update-key "dt,ip" \
--input-null-string '\\N'    \
--input-null-non-string '\\N'
```

定时调度

Azkaban

1. 版本配置文件 编写azkaban.project文件

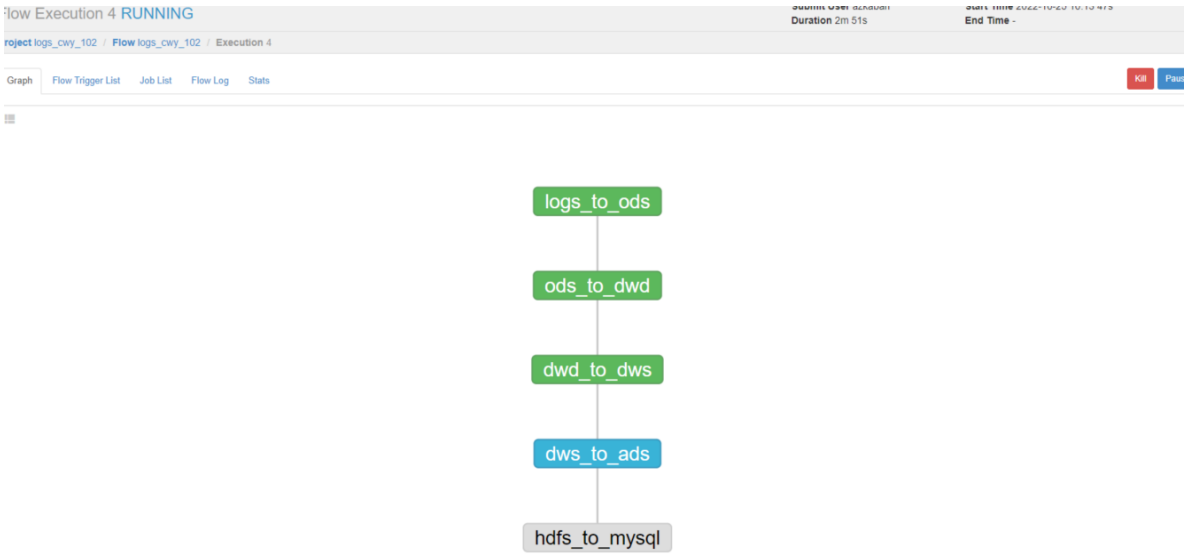
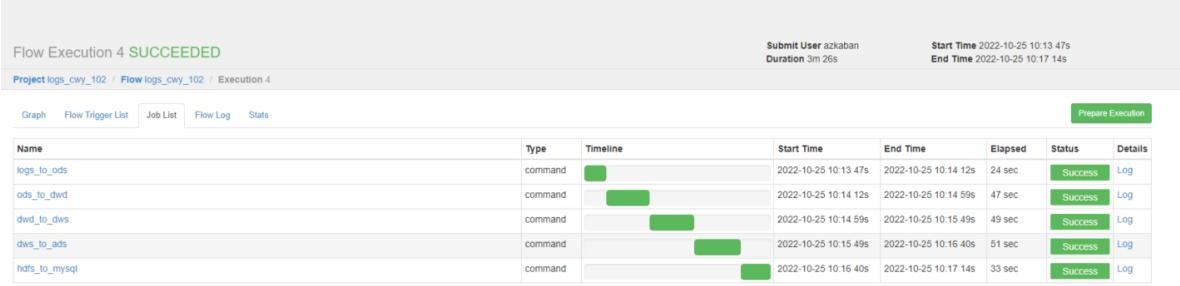
```
azkaban-flow-version: 2.0
```

2. 流程配置文件 logs_cwy_102.flow

```yaml
nodes:
  - name: logs_to_ods
    type: command
    config:
     command: /opt/mock/cwy/logs_cwy/logs_to_ods.sh ${dt}

  - name: ods_to_dwd
    type: command
    dependsOn:
     - logs_to_ods
    config:
     command: /opt/mock/cwy/logs_cwy/ods_to_dwd.sh all ${dt}

  - name: dwd_to_dws
    type: command
    dependsOn:
     - ods_to_dwd
    config:
     command: /opt/mock/cwy/logs_cwy/dwd_to_dws.sh all ${dt}

  - name: dws_to_ads
    type: command
    dependsOn:
     - dwd_to_dws
    config:
     command: /opt/mock/cwy/logs_cwy/dws_to_ads.sh all ${dt}

  - name: hdfs_to_mysql
    type: command
    dependsOn:
     - dws_to_ads
    config:
     command: /opt/mock/cwy/logs_cwy/hdfs_to_mysql.sh all ${dt}
```

3. 打成zip包 上传

注意事件：（多executor模式）

```
提交方案：
1. 指定Executor执行任务
   - 在Mysql的azkaban库中查询executors表，查看master上的Executor的id
   - 在执行工作流程时加入useExecutor属性，useExecutor=id

2. 把Executor所在的所有节点部署任务所需的脚本和应用 （随机选一台执行）
   -  需要分发脚本、sqoop、spark、环境变量（刷新）
   - 重启Azkaban
```

```
dt=2020-06-15 （如果每日定时调度，变量有，但不传值，脚本自动计算前一天的值）
2. 指定execute节点
    useExecutor=id
3. 点击execute执行即可
```